

Optimal Formation Navigation of Multirobot Systems with Layered Obstacles Avoidance

1st Gaolei Yu
School of Mathematics
Southeast University
Nanjing, China
gl.yu@seu.edu.cn

2nd Zhengyang Chen
School of Mathematics
Southeast University
Nanjing, China
zhengyang_chen@seu.edu.cn

3rd Xiaoxiao Li
School of Mathematics
Southeast University
Nanjing, China
30015002lxx@seu.edu.cn

4th Dunle Fang
School of Mathematics
Southeast University
Nanjing, China
java_ding@seu.edu.cn

5th Jingyi Huang
School of Mathematics
Southeast University
Nanjing, China
seujingyi_huang@seu.edu.cn

6th Qingshan Liu*
School of Mathematics
Southeast University
Nanjing, China
qslu@seu.edu.cn

Abstract—This paper investigates the problem of optimal motion planning for multi-robot systems operating in obstacle-dense space while maintaining a predefined formation. We propose a complete operation frame that ranges from global planning to local planning, and from external obstacle avoidance to internal obstacle avoidance. Especially, in our frame, the path planning and formation-keeping are addressed together. Firstly, an enhanced A* algorithm is employed for global path planning. A novel optimization framework based on the concept of Local Free Space is then proposed to determine optimal target positions for the robots. Then, formation maintenance is formulated as a linearly constrained optimization problem through shape theory, and an algorithm is developed for optimal solution. To prevent inter-robot collisions, the reciprocal velocity obstacles method is integrated for local motion coordination. At last, simulation results demonstrate the effectiveness of the proposed approach in achieving collision-free and formation-preserving navigation in complex environments.

Index Terms—multirobot systems, optimal formation, path planning, layered obstacle avoidance

I. INTRODUCTION

Multirobot systems (MRSs) are increasingly important in applications such as logistics, transportation, and search and rescue, where cooperative behavior enhances efficiency, robustness, and scalability. A fundamental challenge is enabling robots to navigate complex environments safely and collaboratively while maintaining formation. Collision-free path planning has therefore become a central research problem, requiring methods that integrate global navigation, local obstacle avoidance, and formation preservation. In the above-mentioned field, the A* algorithm is commonly employed for global planning, though it often generates paths near obstacles [1], [2]. Thus an enhanced variant is proposed in [3] which improves safety and efficiency. Thus, the enhanced A* is applied in the rough global path planning of our research.

After that, in this paper, we investigate the optimization-based approach to achieving optimal motion of a multirobot system with a prescribed formation operating in an obstacle-laden environment. Recent studies have shown that Voronoi-based representations greatly enhance navigation safety and efficiency. The Voronoi Diagram based Collision-free A* Algorithm uses Voronoi topology for adaptive re-planning around dynamic obstacles [4], while the fusion of weighted Voronoi diagrams with improved A* search prioritizes high-clearance, short paths with faster computation [5]. The Clean Generalized Voronoi Diagram (GVD) further improves navigation by pruning redundant edges and generating compact topological graphs for efficient path computation [6]. In addition, the use of power diagrams has been shown to enable exact reactive navigation by defining convex collision-free regions and generating continuous vector fields that guide robots safely to their targets [7]. These approaches highlight the versatility of Voronoi structures [8], but most remain geometric and do not address the dual challenge of obstacle avoidance and formation maintenance. In this paper, to ensure collision avoidance with obstacles, the concept of local free space $\mathcal{LF}$ is employed refer to [7]. Moreover, to preserve the prescribed formation, shape theory is commonly adopted as a strategy [9]. Thus, in this paper, we formulate the optimization problem with the two constraints mentioned above. Actually, the work of [10] and [11] is inspirational to our research. [10] focuses on the optimal formation and optimal matching algorithm with shape theory. And the work of [11] is about the same problem by distributed optimization with Laplacian Matrix. But both [10] and [11] have not considered the environment with obstacles. Moreover, the work of them just focus on the process of forming formation rather than the whole process of path navigation with predefined shape. In our paper, the whole process including navigation and formation-keeping is addressed.

After obtaining the optimal solution, the multi-robot system

*Corresponding author.

must maintain collision-free coordination in dynamic environments. Enhanced Reciprocal Velocity Obstacles (RVO) methods have advanced decentralized motion safety. The V-RVO framework integrates buffered Voronoi cells with RVO cones for dense multirobot avoidance [12], and the modified Hybrid RVO adapts to kinodynamic constraints without inter-agent communication [13]. Meanwhile, Emotional RVO and Level-of-Priority RVO approaches improve leader-follower behaviors by regulating inter-agent distances and avoidance priorities [14], [15]. These developments collectively strengthen both external and internal obstacle-avoidance mechanisms in multirobot formation navigation. These methods achieve scalable collision avoidance, yet they typically focus on local interactions rather than integrating formation-keeping under external obstacles. Thus our paper build a hierarchical obstacle avoidance system, using the method to prevent internal collisions among robots.

The main contribution of the paper is the proposition of a novel framework that jointly addresses path planning and formation-maintaining. By incorporating the notion of local free space, we extend the application of the optimal formation algorithm from formation in empty space to navigation and formation-keeping in obstacle-dense space. And the another contribution is building an internal and external layered obstacle avoidance system. In the external layer, the local free space is applied. And in the internal layer, the *RVO* method is introduced. Thus, this system significantly reduces the complexity of the algorithm while achieving complete obstacle avoidance. The simulation is presented to verify the accuracy and efficiency of the proposition.

Notation: In this paper, $\mathbb{R}$ represents the real number field. And the $\mathbb{R}^d$ means d -dimensional vector space on the $\mathbb{R}$. Especially, the form as $[a, b]^d$ denotes that each dimension of the element inside is constrained within $[a, b]$. Then the $\mathbb{R}^{m \times n}$ represents the m -by- n matrix on $\mathbb{R}$. The $SO(d)$ represents the Special Orthogonal Group of degree d . And the $[S]$ denotes the equivalence class of formations having the same shape as S . Moreover, the $\|\cdot\|$ means the Euclidean norm in the follows.

II. GLOBAL PATH PLANNING

With reference to the seminal paper [16], the cost function of the original A* algorithm is defined by the following core formula:

$$\begin{aligned} f(n) &= g(n) + h(n), \\ g(n) &= g(n-1) + d_{n-1 \rightarrow n}, \end{aligned} \quad (1)$$

where $g(n)$ denotes the actual cost from the start node to the current node n , and $h(n)$ represents the estimated cost from node n to the goal, also referred to as the heuristic function, $f(n)$ represents the total estimated cost of the path through node n . The term $d_{n-1 \rightarrow n}$ corresponds to the actual Euclidean distance from the previous node $n-1$ to node n , which constitutes the base geometric cost.

Building upon the standard A* algorithm, a *safety-aware cost calculation mechanism* has been integrated. Specifically,

the movement cost is modified by multiplying the original cost function by a weight factor W . The formal expression of this cost calculation is defined as follows:

$$\begin{aligned} g(n) &= g(n-1) + W \cdot d_{n-1 \rightarrow n}, \\ W &= 1 + \frac{1}{d_{\min} + \varepsilon}, \end{aligned} \quad (2)$$

where $d_{\min}$ indicates the minimum allowable distance to obstacles or environmental boundaries, and ε is a small constant introduced to avoid division by zero in the cost computation. Through this weighted formulation, paths traversing in close proximity to obstacles or boundaries are penalized, thereby promoting the generation of trajectories with enhanced safety margins.

The A* algorithm is employed to perform a search within a predefined environmental map, generating a globally optimal path $\mathcal{P}$ from the start point B to the target point T^0 . The resulting path comprises an ordered sequence of intermediate waypoints, expressed as $\mathcal{P} = \{B, T^1, T^2, \dots, T^k, T^0\}$. Each robot is guided to traverse these waypoints sequentially, ultimately reaching the designated target location.

III. LOCAL PATH PLANNING

A. Local Free Space

Firstly, this paper considers a closed space containing disc-shaped obstacles $\mathcal{W}$. There are m obstacles and n robots, where the center of each obstacle is denoted by O_j ($1 \leq j \leq m$) and its radius by R_j . The center of each disc-shaped robot is denoted by p_i ($1 \leq i \leq n$), with radius r_i , and the maximum robot radius is defined as $r_m = \max_{1 \leq i \leq n} r_i$. The center of the robot formation is denoted by p_c . To prevent the formation from leaving the workspace, we define an *effective working space* improved from [7] as follows.

$$\widetilde{\mathcal{W}} = \{q \in \mathcal{W} \mid \overline{B(q, r_m)} \in \mathcal{W}\}. \quad (3)$$

For convenience, $\mathcal{W}$ hereafter denotes $\widetilde{\mathcal{W}}$. To prevent collisions, we define the local free space $\mathcal{LF}(x)$ for the robots as follows:

$$\mathcal{LF}(x) = \{q \in \mathcal{W} \mid q \in \bigcap_{i=1}^N \mathcal{H}_i(x)\}. \quad (4)$$

Here, $\mathcal{H}_i$ denotes the half-plane formed by the center of robots p_c and obstacle O_i , defined as

$$\mathcal{H}_i(x) = \{q \in \mathcal{W} \mid (q - p_c)^\top d_i \leq c_i\}, \quad (5)$$

where $d_i = \frac{O_i - p_c}{\|O_i - p_c\|}$, $c_i = \|d_i\| - (R_i + r_m)$.

Then we extend the construction of the half-plane $\mathcal{H}$ to convex polygon obstacles, which are also defined as

$$\mathcal{H}_k(x) = \{q \in \mathcal{W} \mid (q - p_c)^\top v_k \leq c_k\}, \quad (6)$$

where v_k is the unit normal vector of the edge of O_k that is closest to p_c ¹, $d_{Edge,k}$ is the Euclidean distance between p_c

¹If multiple edges are equally close, we select the one whose normal vector forms the largest angle with the current direction of motion.

and this edge, and $c_k = d_{Edge,k} - r_m$. Each obstacle edge is treated as a line segment. Thus, the complete definition of $\mathcal{LF}$ is

$$\mathcal{LF} = \{q \in \mathcal{W} \mid q \in \bigcap_{i=1}^{N_{disk}} \mathcal{H}_{disk,i} \cap \bigcap_{k=1}^{N_{poly}} \mathcal{H}_{poly,k}\}. \quad (7)$$

Since $\mathcal{LF}$ is formed by the intersection of multiple half-planes $\mathcal{H}_i$, thus $\mathcal{LF}$ is convex. Moreover, because of the margin between $\mathcal{LF}$ and obstacles, the whole robots cannot collide with the obstacles when they move in $\mathcal{LF}$.

B. Problem Formulation of Optimal Formation

Then we consider how the multirobot system moves optimally to a target while maintaining the same shape as a given formation, where the formation is defined by n points $s_1, s_2, \dots, s_n$ with $s_i \in \mathbb{R}^2$. Previously, we provide a series of virtual targets $T^1, T^2, \dots, T^l$ along with the real target T^0 to guide the movement using the A* algorithm. Each T^i can be expanded to $(T_1^i, T_2^i, \dots, T_n^i)$ according to the reference formation, where $T_c^i = \frac{1}{n} \sum_{j=1}^n T_j^i$ denotes the centroid of the formation. However, we require that each robot's movement remains within $\mathcal{LF}$. Therefore, we need to determine intermediate waypoints q^k such that the next set of targets lies entirely within $\mathcal{LF}$.

Without loss of generality, we focus on a single action for analysis firstly. To minimize the distance between robots and their targets, the waypoint selection problem is formulated as the following optimization problem:

$$\begin{aligned} \min_Q \quad & \sum_{i=1}^n \frac{1}{2} \lambda \|q_i - T_i\|^2 + (1 - \lambda) \|q_i - T_i\|_1, \\ \text{subject to} \quad & Q \in [S], \\ & q_i \in \mathcal{LF}, i = 1, 2, \dots, n, \end{aligned} \quad (8)$$

where $\lambda \in [0, 1]$ is a weight coefficient to adjust the compactness of the formation, $q_i \in \mathbb{R}^2$ denotes the goal position of robot i , $Q = (q_1, q_2, \dots, q_n) \in \mathbb{R}^{2 \times n}$ is the target formation matrix, $S = (s_1, s_2, \dots, s_n) \in \mathbb{R}^{2 \times n}$ represents the reference formation, and T is the current target for the group of robots.

To simplify the constraint, we transform $Q \in [S]$ into a linear equality constraint $Aq = 0$ [9], where $q = (q_1^\top, q_2^\top, \dots, q_n^\top) \in \mathbb{R}^{2n}$, $M_i = \begin{pmatrix} s_{i,x} & -s_{i,y} \\ s_{i,y} & s_{i,x} \end{pmatrix}$ and

$$A = \begin{pmatrix} M_3 - M_2 & -M_3 & M_2 & O & \dots & O \\ M_4 - M_2 & -M_4 & O & M_2 & \dots & O \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ M_n - M_2 & -M_n & O & O & \dots & M_2 \end{pmatrix}.$$

Lemma 1. The formula (8) is rewritten as

$$\begin{aligned} \min_Q \quad & \frac{1}{2} \lambda \|q - T\|^2 + (1 - \lambda) \|q - T\|_1, \\ \text{subject to} \quad & Aq = 0, \\ & q \in \mathcal{LF}^n, \end{aligned} \quad (9)$$

where $T = (T_1^\top, T_2^\top, \dots, T_n^\top) \in \mathbb{R}^{2n}$ and $\mathcal{LF}^n = \prod_{i=1}^n \mathcal{LF}$.

Proof. $Q \in [S]$ means Q can be obtained from S only through rotation, scaling, and translation. Let $s_1 = (0, 0)$, we have $p_i - p_1 = cRs_i$ where positive constant c presents scaling factor and matrix $R \in SO(2)$ presents rotation factor. We consider vector p_i and s_i as complex number. Thus R also mean a complex number denoted r . Then we have a formula about complex number $p_i - p_1 = crs_i$. When $i = 1$, it is $p_2 - p_1 = crs_2$. Dividing the two formula yields $\frac{p_i - p_1}{p_2 - p_1} = \frac{s_i}{s_2}$, namely $s_2 \cdot (p_i - p_1) = s_i \cdot (p_2 - p_1)$. The above equation can be rewritten in matrix form as $M_2(p_i - p_1) = M_i(p_2 - p_1)$, which organized to $(M_i - M_2)p_1 - M_i p_2 + M_2 p_i = 0$. Thus, the strengthened proof has been presented. $\square$

Every action of multirobot system follow the same logic as above.

IV. OPTIMAL FORMATION ALGORITHM DESCRIPTION AND CONVERGENCE ANALYSIS

In this section, we find the optimal solution of the optimization problem in (9). The following lemma provides the necessary and sufficient conditions for this solution, which proof is similar to Theorem 1 in [17].

Lemma 2. [17] Assume σ to be a positive constant, $\phi : \mathbb{R}^{nm} \rightarrow \mathcal{LF}^n$ and $\psi : \mathbb{R}^{nm} \rightarrow [-1, 1]^{nm}$ to be projection operators. $q^* \in \mathbb{R}^{nm}$ is an optimal solution of (9), if and only if there exist $y^* \in \mathbb{R}^l$ (l is the number of rows of A) and $z^* \in \mathbb{R}^{nm}$ such that (q^*, y^*, z^*) satisfies

$$\begin{cases} q^* = \phi(q^* - \sigma(\lambda(q^* - p) + (1 - \lambda)z^* + A^\top y^*)), \\ Aq^* = 0, \\ z^* = \psi(z^* + q^* - p). \end{cases} \quad (10)$$

Then we present an iterative algorithm to find the solution of the equations in (10), as follows:

$$\begin{cases} q_{k+1} = \phi(q_k - \sigma(\lambda(q_k - p) + (1 - \lambda)\psi(z_k + q_k - p) + A^\top(y_k + Aq_k))), \\ y_{k+1} = y_k + Aq_{k+1}, \\ z_{k+1} = \psi(z_k + q_{k+1} - p). \end{cases} \quad (11)$$

Here we use the subscript to simplify the iterations, i.e., $q_k = q(k)$, $y_k = y(k)$, $z_k = z(k)$. The first formula represents the descent with a constant step length gradient and formation-maintenance. The y_k is a Lagrange multiplier about that constraint $Aq = 0$. And the z_k is a subgradient auxiliary variable of L1-norm, in order to handle the unsmooth function L1-norm.

Theorem 1. For any initial condition (q_0, y_0, z_0) , the iterative sequence $\{q_k\}$ generated by the projection-based algorithm converges globally to an optimal solution q^* of the formation optimization problem, provided that the step-size parameter σ satisfies $\sigma < \frac{1}{2 + \lambda_{\max}(A^\top A)}$, where $\lambda_{\max}(\cdot)$ denotes the maximum eigenvalue of the matrix $A^\top A$, and A is the full row-rank matrix constructed from the formation constraints.

Proof. The convergence proof follows the methodology established in [10]. Consider the Lyapunov candidate function defined as:

$$V(q_k, y_k, z_k) = \|q_k - q^*\|^2 + \sigma (\|y_k - y^*\|^2 + (1 - \lambda) \|z_k - z^*\|^2).$$

The convergence properties are established through the following inequalities derived from the update rules:

1) The incremental change in $V_1(q_k) = \|q_k - q^*\|_2^2$ satisfies:

$$\begin{aligned} V_1(q_{k+1}) - V_1(q_k) &\leq -\|q_{k+1} - q_k\|_2^2 \\ &\quad + 2\sigma\lambda(q_{k+1} - q_k)^\top (q_{k+1} - q_k) \\ &\quad - 2\sigma(1 - \lambda)(q_{k+1} - p)^\top (\psi(z_k + q_k - p) - z^*) \\ &\quad - 2\sigma(Aq_{k+1})^\top (y_k - y^* + Aq_k). \end{aligned}$$

2) The update of the dual variable yields:

$$\begin{aligned} V_2(y_{k+1}) - V_2(y_k) &= 2(Aq_{k+1})^\top (y_k - y^* + Aq_k) \\ &\quad + \|A(q_{k+1} - q_k)\|_2^2 \\ &\quad - \|Aq_k\|_2^2. \end{aligned}$$

3) The change in the subgradient term satisfies:

$$\begin{aligned} V_3(z_{k+1}) - V_3(z_k) &\leq -\|z_{k+1} - z_k\|_2^2 \\ &\quad + 2(q_{k+1} - p)^\top (z_{k+1} - z^*). \end{aligned}$$

By combining these inequalities, the total Lyapunov function $V(q_k, y_k, z_k)$ is non-increasing along the system trajectories.

According to the LaSalle invariance principle for discrete-time systems, and analogous to the proof presented in reference [10], it can be rigorously established that the algorithm achieves global convergence. $\square$

After the solution is obtained, the robots move to the target position using the *RVO* algorithm, which is described in the next section.

V. INTERNAL OBSTACLE AVOIDANCE

In multi-robot formation control, inter-robot collision avoidance is critical. Internal collision avoidance is implemented based on the *RVO* method described in reference [18]. Formally, for robots i and j with velocities $v_i(t)$ and $v_j(t)$, the *RVO* imposed on i by j is defined as:

$$RVO_j^i(v_j(t), v_i(t)) = \{v'_i(t) \mid 2v'_i(t) - v_i(t) \in VO_j^i(v_j(t))\},$$

where, $VO_j^i(v_j(t))$ represents velocities of i that would result in a collision with j assuming $v_j(t)$ is constant. Velocities within the *RVO* set are weighted averages between current velocity and a VO-derived velocity, ensuring reciprocal collision avoidance.

In multi-robot systems, each robot first computes its preferred velocity $v_{pref,i}(t)$ and checks for conflicts with other robots' *RVO* sets. If a conflict exists, an optimization problem selects a velocity that avoids all *RVO* sets while minimizing deviation from $v_{pref,i}(t)$:

$$v_i(t) = \arg \min_{v \in G_i(t) \setminus O_i(t)} \|v - v_{pref,i}(t)\|,$$

where $G_i(t)$ is the set of reachable velocities under kinematic and dynamic constraints, and $O_i(t)$ is the union of all velocity obstacles from other robots and static obstacles. If no fully collision-free velocity exists $G_i(t) \setminus O_i(t) = \emptyset$, the time-to-collision $t_i^{col}(v, t)$ is used as a penalty, and a suboptimal velocity is chosen by:

$$v_i(t) = \arg \min_{v \in G_i(t)} \left(\frac{\beta}{t_i^{col}(v, t)} + \|v - v_{pref,i}(t)\| \right),$$

where $\beta > 0$ balances safety and tracking performance.

Algorithm 1 Optimal Formation Navigation Algorithm

Initialization:

- 1: Set the start points B , target point T^0 , reference shape S and step size σ .
- 2: Calculate A and σ .
- 3: Initialize global path $\mathcal{P}$ by *A*Enhanced* in (1) and (2).

Iteration:

- 4: Get the first target $T = T^1$.
 - 5: **while** current position $P \neq T^0$ **do**
 - 6: **for** $k = 1$: Maximum number of iterations **do**
 - 7: Using the iterations in (11) to get the optimal waypoints Q^* .
 - 8: **end for**
 - 9: Update P to Q^* .
 - 10: **if** $Q^* = T$ **then**
 - 11: Set T as the next target T^i .
 - 12: **else**
 - 13: Update $\mathcal{P}$ by inserting Q^* before T .
 - 14: **end if**
 - 15: **end while**
 - 16: **return** the whole path $\tilde{\mathcal{P}}$ updated.
-

VI. SIMULATION RESULTS

A $[0, 20] \times [0, 20]$ planar workspace with static disk-shaped obstacles was configured. Four homogeneous robots ($n = 4$) were deployed in initial square formation, with virtual reference center q_c at the origin. The proposed algorithm was compared against a baseline A* planner with fixed formation geometry. Than the multirobot system will action through Algorithm 1.

Figure 1 demonstrates successful navigation through the constrained environment. All robots reached targets within 45 seconds while maintaining collision avoidance despite dense obstacles. The algorithm navigated five narrow passages with consistent safety margins 0.3, maintaining computational efficiency (average 12 ms/step optimization).

Robots trajectories show effective navigation while preserving formation integrity. Adaptive safety regions $\mathcal{LF}(x)$ (green) enable safe corridor passage by constraining formation movement based on obstacle proximity. The path comprises 12

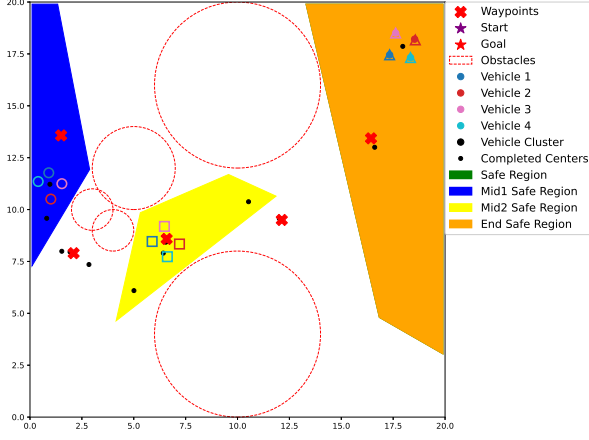


Fig. 1. Multi-robot navigation trajectories in obstacle-dense environment. Purple/red stars mark start/goal positions, black lines show paths. Green polygons $\mathcal{LF}(x)$ represent dynamic safety regions, red circles indicate obstacles.

A*-generated waypoints, with safety projection onto $\mathcal{LF}(x)$ applied when necessary.

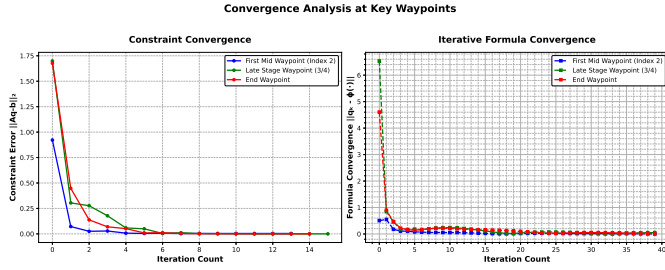


Fig. 2. Convergence analysis at navigation waypoints. (Left) Constraint convergence via $\|Aq - b\|$. (Right) Behavior of iterative update $\|q_k - \phi(q_k - \sigma(\lambda(q_k - p) + (1 - \lambda)z_k + A^T y_k))\|$.

Figure 2 shows comprehensive convergence analysis. The left plot confirms strict formation constraint satisfaction ($Aq = b$), with residual norms rapidly decreasing below threshold to machine precision (10^{-14}). This ensures geometric integrity, with most waypoints converging within 20 iterations. The right plot proves that the optimization function reaches the minimum value, through plotting the iterative updated $\|q_k - \phi(q_k - \sigma(\lambda(q_k - p) + (1 - \lambda)z_k + A^T y_k))\|$ and finding it limits to 0.

VII. CONCLUSION

This paper's proposition has achieved certain accomplishments in accuracy, convergence speed and *etc.*. In the future, the focus of work will be on the design of distributed algorithm and hardware implementation.

REFERENCES

- [1] Y. Yang and Q. Cai, "Research on the a-star algorithm based on path finding," in *2023 IEEE 3rd International Conference on Data Science and Computer Application (ICDSCA)*. IEEE, 2023, pp. 199–202.
- [2] M. Li and H. Zhang, "Auv 3d path planning based on a* algorithm," in *2020 Chinese automation congress (CAC)*. IEEE, 2020, pp. 11–16.
- [3] B. Cao, Z. Yang, L. Yu, and Y. Zhang, "Research on the star algorithm for safe path planning," in *2023 IEEE International Conference on Control, Electronics and Computer Technology (ICCECT)*. IEEE, 2023, pp. 105–109.
- [4] S.-L. Ho, J.-K. Lin, K.-Y. Chou, and Y.-P. Chen, "Voronoi diagram based collision-free a* algorithm for mobile vehicle in complex dynamic environment," in *2022 IEEE International Conference on Consumer Electronics-Taiwan*. IEEE, 2022, pp. 429–430.
- [5] Z. Liu, L. Gao, F. Liu, D. Liu, and W. Han, "Fusion of weighted voronoi diagram and a* algorithm for mobile robot path planning," in *2022 2nd International Conference on Electrical Engineering and Mechatronics Technology (ICEEMT)*. IEEE, 2022, pp. 403–406.
- [6] H. Tian, Z. Tang, Z. Zhang, H. Deng, Z. Pei, and P. Jiang, "Clean generalized voronoi diagram: An efficient algorithm for path planning in robotics," in *2025 IEEE International Conference on Mechatronics and Automation (ICMA)*. IEEE, 2025, pp. 759–763.
- [7] O. Arslan and D. E. Koditschek, "Exact robot navigation using power diagrams," in *2016 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2016, pp. 1–8.
- [8] K. Sugihara, "Why are voronoi diagrams so fruitful in application?" in *2011 Eighth International Symposium on Voronoi Diagrams in Science and Engineering*. IEEE, 2011, pp. 14–14.
- [9] E. Montijano and A. R. Mosteo, "Efficient multi-robot formations using distributed optimization," in *53rd IEEE Conference on Decision and Control*. IEEE, 2014, pp. 6167–6172.
- [10] Q. Liu and M. Wang, "A projection-based algorithm for optimal formation and optimal matching of multi-robot system," *Nonlinear Dynamics*, vol. 104, no. 1, pp. 439–450, 2021.
- [11] J. Huang, S. Zhou, H. Tu, Y. Yao, and Q. Liu, "Distributed optimization algorithm for multi-robot formation with virtual reference center," *IEEE/CAA Journal of Automatica Sinica*, vol. 9, no. 4, pp. 732–734, 2022.
- [12] S. H. Arul and D. Manocha, "V-rvo: Decentralized multi-agent collision avoidance using voronoi diagrams and reciprocal velocity obstacles," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2021, pp. 8097–8104.
- [13] M. Sainte Catherine and E. Lucet, "A modified hybrid reciprocal velocity obstacles approach for multi-robot motion planning without communication," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2020, pp. 5708–5714.
- [14] M. Fachri, M. Hariadi, and S. M. S. Nugroho, "Emotional reciprocal velocity obstacles for leader-following: A velocity obstacles based steering behavior," *IEEE Access*, 2024.
- [15] M. Fachri, S. Juniastuti, S. M. S. Nugroho, and M. Hariadi, "Level of priority based leader-following behavior using reciprocal velocity obstacles in multi-agents navigation," in *2021 International Seminar on Machine Learning, Optimization, and Data Science (ISMODE)*. IEEE, 2022, pp. 45–50.
- [16] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.
- [17] Q. Liu and K. Li, "A continuous-time algorithm based on multi-agent system for distributed least absolute deviation subject to hybrid constraints," in *IECON 2017-43rd Annual Conference of the IEEE Industrial Electronics Society*. IEEE, 2017, pp. 7381–7386.
- [18] Y. Hu, H. Yu, Y. Zhong, and Y. Lv, "Distributed collision-avoidance formation control: A velocity obstacle-based approach," in *2019 IEEE Symposium Series on Computational Intelligence (SSCI)*. IEEE, 2019, pp. 1994–2000.